Bayesian Network

It is a graphical directed model which represents certain independent/conditional relationships between the different variables in the domain.

This helps to tractable inference in many cases where the variables are not completely connected or completely dependent on each other.

Variables $X_1, X_2, \ldots X_n, Y$
- Look for types of dependences or other independences between them
- represented through nodes

Arcs — defines relations between nodes

Nodes and Arcs represents conditional dependence, conditional independence relationships

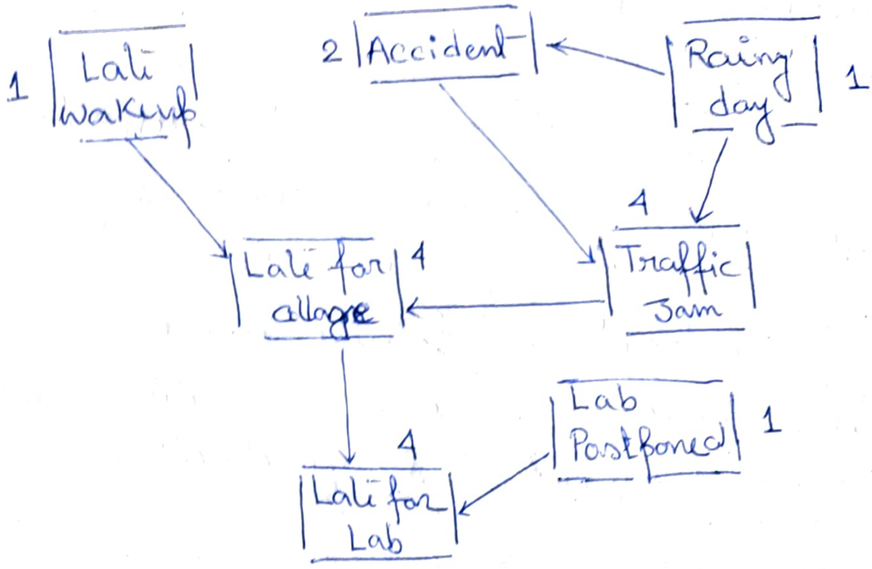
Hence Bayesian Network represents causality

* causality help to reason and take decision like human brain

It is due to one node is the cause of other node or effect of other node.

When these causality is represented that forms the Bayesian network structure.

Bayesian network can also be represented without causality but the efficiency of the network will be less.



Directed acyclic graph (DAG)

The arcs over here indicates causal relationships between the nodes. — also can read some conditional independence relationship

'Late wakeup' and 'Late for Lab' are not independent but they are conditionally independent as being waking up late, the person may not be late for college.

Represents efficiently joint probability distribution of the variable.

* Joint probability distribution: given two random variables that are defined on the same probability space, the joint probability distribution is the probability is the corresponding probability distribution of all possible pairs of outputs.

If the graph is having less no. of edges then the variables are conditionally independent of others and joint distribution can be represented more efficiently.

Arc — Probabilistic dependence among variables

Absence of arc denotes independence or conditional or conditional independence

Let, the 7 variables are Boolean so 2^7 possible combination

Keep probability distribution of node given the value of its parents

No parent → node → 1

accident has 1 parent → rainy day

probability of accident being true given that it is a rainy day and accident being true that it is not a rainy day

So 1 parent → need to keep 2 values.

Thus total number of probability values that have to be kept 17 probability values in contrast of (2^7-1)

— So Bayesian Network is the compact representation of joint probability distribution.

Bayesian Network is used to draw Inference.

Inference is used to compute posterior probability given about some nodes.

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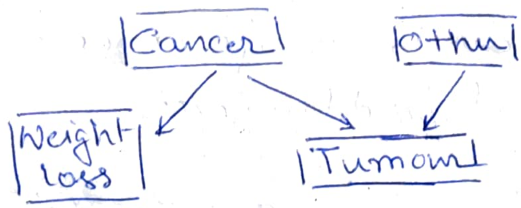
Application – Diagnosis



Hence from symptoms try to find the probability of cancer – Hence used for prediction

Diagnosis: $P(\text{Cause} \mid \text{Symptom}) = ?$

Prediction: $P(\text{Symptom} \mid \text{Cause}) = ?$

Classification: $P(\text{Class} \mid \text{Data}) = ?$

Decision making

Structure of the graph $\Leftrightarrow$ conditional independence relations

$$P(x_1, x_2, \ldots x_n) = \underset{\text{Product of}}{\prod} P(x_i \mid \text{parent}(x_i))$$

$$P(x_1, x_2, \ldots x_n) = P(x_1)\, P(x_2 \mid x_1)\, P(x_3 \mid x_1 x_2)$$

↳ normal chain rule for joint probability distribution.